

Quantum Algorithm Performance Under Noise

Yuvraj

Department of Computer Science
University of Maryland-College Park
ymehta13@terpmail.umd.edu

Abstract

This study examines how noise affects the reliability of quantum circuits through two complementary approaches: a controlled simulation using Qiskit Aer and an execution study on real IBM Quantum hardware. In simulation, five noise models (bit-flip, phase-flip, depolarizing, readout, and thermal-relaxation) were applied to a parameterized benchmark circuit across varying qubit counts, circuit depths, and noise strengths, producing 6,000 experimental observations. Results showed that success probability declined substantially as qubit count and circuit depth increased, with bit-flip, phase-flip, and depolarizing noise degrading performance far more severely (mean success probabilities of 0.597, 0.580, and 0.607, respectively) than readout noise (0.917) or thermal-relaxation noise (0.982). The same benchmark circuit was then executed on the `ibm_marrakesh` processor across 150 hardware runs (150,000 total shots). Hardware performance remained consistently high, with an overall mean success probability of 0.992, showing a measurable decline with increasing qubit count (0.998 at two qubits to 0.986 at six) but comparatively little sensitivity to circuit depth. These results indicate that controlled noise mechanisms and real hardware degrade quantum circuit performance through different underlying dynamics, with implications for how simulated noise models should be interpreted relative to physical device behavior.

1 Introduction

Quantum computers have the potential to solve certain computational problems in ways that differ fundamentally from classical computers. However, current quantum processors are highly susceptible to noise and operational errors. Imperfections in quantum gates, measurement processes, and qubit coherence can cause the state produced by a quantum circuit to deviate from its ideal result. As quantum circuits become larger and deeper, understanding how these errors accumulate is an important part of evaluating the reliability of current quantum computing systems.

This project investigates the effect of noise on quantum circuit performance through two complementary experimental approaches. First, a controlled simulation study was conducted using Qiskit Aer to examine how different noise mechanisms affect circuit reliability under systematically varied conditions. Five noise models were considered: bit-flip, phase-flip, depolarizing, readout, and thermal-relaxation noise. The experiments varied the number of qubits, circuit depth, and noise strength while measuring the resulting success probability and measurement-distribution fidelity.

Second, the study was extended from simulation to real quantum hardware using an IBM Quantum processor. The same benchmark circuit structure was executed on an IBM QPU across multiple circuit sizes and depths, with repeated measurements used to characterize the variability and error observed on actual hardware. This provides a comparison between controlled noise simulations and the behavior of a physical quantum processor.

The primary objective of the study is to characterize how circuit reliability changes as quantum workloads become more complex and as noise increases. By combining controlled simulations with real-hardware measurements, the project provides an experimental framework for studying the relationship between circuit structure, noise, and quantum computation reliability.

2 Background

2.1 Quantum Noise

Quantum noise refers to processes that cause the state of a quantum system to deviate from its ideal evolution. In a quantum circuit, these deviations can occur during gate operations, while qubits evolve over time, or during measurement. The effect of noise becomes increasingly relevant as circuits contain more operations or involve more qubits.

This study models several noise mechanisms independently to examine how each affects the output of the same benchmark circuit. The simulated noise models are bit-flip, phase-flip, depolarizing, readout, and thermal-relaxation noise.

2.2 Noise Models

Bit-flip noise applies an X error with probability p , changing a qubit from $|0\rangle$ to $|1\rangle$ or vice versa.

Phase-flip noise applies a Z error with probability p . Unlike a bit flip, a phase flip changes the relative phase of quantum states without directly changing computational-basis measurement values.

Depolarizing noise represents a more general error process in which the quantum state is partially randomized. In the experiments, depolarizing errors were applied to both single- and two-qubit operations.

Readout noise occurs during measurement rather than during circuit evolution. In the implemented model, each measured bit has probability p of being reported incorrectly.

Thermal-relaxation noise models energy relaxation and dephasing using T_1 and T_2 coherence parameters. In this experiment, $T_1 = 50\ \mu\text{s}$ and $T_2 = 70\ \mu\text{s}$ were used, while the experimental noise-strength parameter controlled the effective gate duration.

2.3 Circuit Depth and Qubit Count

Two structural properties of the benchmark circuit were varied: circuit depth and number of qubits. Circuit depth controls how many repeated computational layers are executed, increasing the number of opportunities for errors to affect the computation. Qubit count controls the size of the quantum system and changes the number of single- and two-qubit operations required by the circuit.

The experiments therefore evaluate noise as a function of three primary variables:

Noise strength, Circuit depth, Number of qubits

This allows the effects of increasing noise intensity to be separated from the effects of increasing circuit size.

2.4 Performance Metrics

The primary metric is success probability, defined as the fraction of measurement shots producing the expected computational-basis state:

$$P_{\text{success}} = \frac{N_{\text{correct}}}{N_{\text{shots}}} \quad (1)$$

The corresponding error rate is:

$$E = 1 - P_{\text{success}} \quad (2)$$

The study also calculates measurement-distribution fidelity between the ideal and noisy output distributions:

$$F(P, Q) = \left(\sum_x \sqrt{P(x)Q(x)} \right)^2 \quad (3)$$

This is a classical fidelity measure between probability distributions and should not be interpreted as a full quantum-state fidelity. Because the benchmark circuit has a deterministic ideal output, the ideal distribution contains all probability mass on a single state, making the measured distribution fidelity equivalent to the success probability for this experiment.

3 Methodology

This study uses a controlled simulation experiment followed by measurements on real quantum hardware. The simulation experiment was designed to isolate the effects of different noise mechanisms while systematically varying circuit size, circuit depth, and noise strength. The hardware experiment used the same benchmark circuit structure to measure the behavior of a physical quantum processor.

3.1 Benchmark Circuit

The benchmark circuit was constructed using Hadamard gates and nearest-neighbor controlled-NOT (CNOT) gates. The circuit begins by applying a Hadamard gate to each qubit, followed by repeated layers of nearest-neighbor CNOT operations and Hadamard gates. A corresponding reverse sequence is then applied to uncompute the state.

The circuit is parameterized by the number of qubits and its depth. The uncomputation produces a deterministic ideal output, allowing the expected measurement state to be identified without ambiguity. This makes deviations from the expected state directly measurable as computational errors.

3.2 Simulation Experiment

The simulated experiments were performed using Qiskit Aer. Five noise models were evaluated independently:

1. Bit-flip noise
2. Phase-flip noise
3. Depolarizing noise
4. Readout error
5. Thermal-relaxation noise

The number of qubits was varied across 2, 3, 4, 5, and 6 qubits. Circuit depth was varied across 1, 2, 4, 8, 16, and 32. For the first four noise models, the noise probability was varied across 0, 0.001, 0.002, 0.005, 0.010, 0.020, 0.050, and 0.100. Thermal-relaxation experiments used the same experimental noise-strength parameter to control the effective gate duration, with $T_1 = 50 \mu s$ and $T_2 = 70 \mu s$.

Each experimental condition was executed for 1,000 measurement shots and repeated five times using independent simulation seeds. The resulting dataset contains 6,000 experimental rows, corresponding to 5 noise models, 5 qubit counts, 6 circuit depths, 8 noise levels, and 5 repetitions.

3.3 Performance Metrics

Circuit performance was evaluated using success probability, error rate, and measurement-distribution fidelity, as defined in Equations (1)–(3). Because the benchmark circuit has a deterministic ideal output, this distribution-based fidelity reduces to the success probability for the experimental conditions used in this study.

3.4 Statistical Analysis

The repeated measurements were used to quantify variation between independent runs. Mean performance, standard deviation, and 95% confidence intervals were calculated for the experimental conditions. Additional aggregation was performed across noise models, qubit counts, circuit depths, and noise strengths to characterize performance trends.

3.5 Real Hardware Experiment

The second stage of the study was conducted on an IBM Quantum processor, `ibm_marrakesh`. The benchmark circuit was transpiled for the hardware before execution to map the circuit onto the processor’s native gate set.

The hardware experiment used 2–6 qubits, circuit depths of 1, 2, 4, 8, 16, and 32, and five repetitions for each condition. Each circuit was executed using 1,000 shots, producing 150 experimental measurements and 150,000 hardware shots in total.

Unlike the simulation experiment, the hardware experiment did not apply an artificial noise model. The measured errors therefore arise from the behavior of the physical quantum processor and its measurement system under the selected circuit conditions.

4 Results

4.1 Simulation Results

The Qiskit Aer experiment produced 6,000 observations across five noise models, five qubit counts, six circuit depths, eight noise strengths, and five repetitions. Across all simulation conditions, the mean success probability was 0.737.

The average success probability varied substantially between noise models. Thermal-relaxation noise produced the highest average success probability at 0.982, followed by readout noise at 0.917. Depolarizing, bit-flip, and phase-flip noise produced lower average success probabilities of 0.607, 0.597, and 0.580, respectively (Table 1).

Success probability also decreased as the number of qubits increased. Averaged across noise models, depths, and noise strengths, mean success probability decreased from 0.826 for two qubits to 0.670 for six qubits (Table 2).

Table 1: Mean success probability and error rate by noise model.

Noise model	Mean success probability	Mean error rate
Bit-flip	0.597	0.403
Phase-flip	0.580	0.420
Depolarizing	0.607	0.393
Readout	0.917	0.083
Thermal relaxation	0.982	0.018

Table 2: Mean success probability by qubit count (simulation).

Qubits	Mean success probability
2	0.826
3	0.767
4	0.725
5	0.694
6	0.670

A similar decrease was observed as circuit depth increased. Mean success probability decreased from 0.864 at depth 1 to 0.601 at depth 32 (Table 3).

Table 3: Mean success probability by circuit depth (simulation).

Depth	Mean success probability
1	0.864
2	0.819
4	0.768
8	0.713
16	0.656
32	0.601

These results show that, within the simulated conditions, increasing circuit depth and qubit count was associated with lower circuit success probability.

4.2 Real Hardware Results

The IBM Quantum experiment produced 150 measurements across the same five qubit counts and six circuit depths, with five repetitions per condition. Each measurement used 1,000 shots, resulting in 150,000 hardware shots.

The overall mean success probability on the IBM hardware was 0.992, corresponding to an average error rate of 0.008.

Mean success probability decreased as the number of qubits increased. The average success probability was 0.998 for two qubits and 0.986 for six qubits (Table 4).

Unlike the simulation results, the average hardware success probability did not show a strictly decreasing relationship with circuit depth over the tested conditions. Mean success probabilities ranged from 0.991 to 0.993 across the six depths (Table 5).

Table 4: Mean success probability by qubit count (IBM hardware).

Qubits	Mean success probability
2	0.998
3	0.996
4	0.990
5	0.989
6	0.986

Table 5: Mean success probability by circuit depth (IBM hardware).

Depth	Mean success probability
1	0.992
2	0.993
4	0.992
8	0.992
16	0.991
32	0.991

The hardware measurements therefore showed a clearer dependence on qubit count than on circuit depth within the tested range.

5 Discussion

The simulation results demonstrate a consistent relationship between increasing circuit complexity and decreasing computational reliability. Across the simulated experiments, success probability decreased as both the number of qubits and circuit depth increased. This behavior is consistent with the increasing number of opportunities for errors to affect the computation as additional qubits and operations are introduced.

The simulated noise models produced substantially different effects. Bit-flip, phase-flip, and depolarizing noise resulted in considerably larger reductions in success probability than readout and thermal-relaxation noise under the parameters used in this study. The differences indicate that the location and structure of an error mechanism within the computation can strongly influence its effect on the final measurement distribution.

Circuit depth had a particularly clear effect in the simulation. Mean success probability decreased from 0.864 at depth 1 to 0.601 at depth 32. This indicates that repeated computational layers accumulated errors over the course of the circuit. Increasing the number of qubits also reduced performance, although the change was smaller than the observed change across circuit depths.

The IBM hardware experiment exhibited a different scale of error. Hardware success probability remained above approximately 98% for the tested conditions, with an overall mean of approximately 99.2%. The effect of qubit count was measurable, with mean success probability decreasing from approximately 99.8% for two qubits to 98.6% for six qubits. However, the dependence on circuit depth was comparatively weak over the tested range.

The difference between the simulated and hardware results should not be interpreted as a direct comparison between an individual simulated noise model and the IBM processor. The simulation applies controlled noise models with predefined parameters, whereas the hardware

experiment contains device-specific sources of error that are not represented by any single model in the simulation. The two experiments therefore serve complementary purposes: the simulation isolates specific noise mechanisms, while the hardware experiment measures the aggregate behavior of a physical processor.

A further distinction is that the thermal-relaxation model used in the simulation is parameterized using assumed T_1 and T_2 values and an effective gate-time scaling. It therefore represents a controlled model of relaxation and dephasing rather than a direct reconstruction of the noise characteristics of the IBM processor.

Overall, the experiments show that circuit size and structure are important factors in quantum circuit reliability. The simulation provides controlled measurements of how specific noise mechanisms affect performance, while the hardware experiment demonstrates how errors manifest on an actual quantum processor.

6 Limitations

Several limitations should be considered when interpreting the results of this study.

First, the benchmark circuit is designed specifically for controlled noise analysis and does not represent a general-purpose quantum algorithm. Its deterministic ideal output simplifies the measurement of errors but does not capture the behavior of algorithms whose outputs are inherently probabilistic.

Second, the simulated noise models are simplified representations of individual error mechanisms. Real quantum processors can exhibit device-specific, time-dependent, and correlated errors that are not fully represented by the independent noise models used in this experiment.

Third, the thermal-relaxation model uses fixed T_1 and T_2 parameters and an effective gate-time scaling. These parameters provide a controlled simulation environment but do not constitute a complete characterization of the IBM processor used in the hardware experiment.

Fourth, the hardware experiment was conducted on a single IBM Quantum processor and a limited range of circuit sizes. The results therefore characterize the tested hardware and conditions rather than quantum hardware in general.

Finally, the study uses 1,000 shots per experimental condition and five repetitions. Increasing the number of shots or repetitions could provide more precise estimates of the measured probabilities and confidence intervals.

Despite these limitations, the combination of controlled simulation and real-hardware measurements provides a consistent framework for examining how circuit size, depth, and noise affect quantum circuit reliability.

7 Conclusion

This study evaluated the effect of noise on quantum circuit reliability using both controlled simulation and real quantum hardware. The simulation experiment evaluated 6,000 conditions across five noise models, multiple qubit counts, circuit depths, and noise strengths. The results showed that increasing circuit depth and qubit count generally reduced success probability, with the magnitude of the effect depending on the noise model.

The real-hardware experiment extended the analysis to 150 measurements on an IBM Quantum processor. Hardware success probability remained high across the tested conditions, while a measurable reduction in performance was observed as the number of qubits increased. The hardware

results also showed that the behavior of a physical processor cannot be represented directly by a single simplified noise model.

The combination of simulation and hardware measurements provides two complementary views of quantum circuit reliability. Controlled simulations allow individual noise mechanisms to be isolated and analyzed systematically, while hardware experiments capture the combined effects of errors present in a physical processor. Together, the results demonstrate the importance of characterizing noise as quantum circuits increase in size and complexity.

Future work could extend the study to additional circuit types, larger qubit counts, additional hardware platforms, and more detailed device-specific noise models. The resulting noise characterization could also be used as a basis for studying quantum error-correction techniques and evaluating how effectively they reduce logical error rates.

References

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